

Set



Agenda

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- ② Creating set object
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- ④ built-in methods
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Set

set is a class

set is mutable

set is not hashable

set is iterable

set is not a sequence

set cannot have duplicate values

indexing is not applicable to set object

slicing operator is not applicable

set does not guarantee to store values
in the order of insertion

How to create set object?

$S1 = \{1, 5, 8\}$

$S2 = \{10, 2, 8, 10, 10, 8\}$

$S3 = \{ \}$ not a set object

$S4 = \text{set}()$

$S5 = \text{set}([1, 5, 3])$

Elements of set object must be hashable

Accessing set elements

for e in setObject:
 code

built-in methods

len()

min()

max()

sum()

sorted()

Concatenation and Repetition Operator

set + set

not supported

set * int

not supported

Comparison Operator

$S1 > S2$

$S1 < S2$

$S1 \geq S2$

$S1 \leq S2$

$S1 == S2$

$S1 != S2$

} Supported

Two set objects are equal if their elements are same, doesn't matter the order of elements.

set object methods

`add()` add specified item in a set

`update()` add iterable(s) elements

`discard()` remove item

`remove()`

`intersection()`

`union()`

`clear()`

`issubset()`

`issuperset()`

`pop()`

Set Comprehension

$S1 = \{ \text{expression for } e \text{ in object} \}$

`s = input("Enter a string")`

`S1 = { e for e in s if e in "aeiou" }`

for e in s:

if e in "aeiou":

`s1.add(e)`

user input

```
S = input("Enter elements separated by comma")
```

```
S1 = {eval(e) for e in S.split(',')}
```

Frozenset

A set is a mutable object while frozenset provides an immutable implementation.